quiz 叁 6Nov

IGCSE Physics 0625 · Thermal Physics

Name: _____ **Class:** _____

Time: 40 minutes **Total: 32 marks**

INSTRUCTIONS

Answer all questions.

Write in dark blue or black pen. You may use an HB pencil for diagrams.

Give non-exact numerical answers correct to 3 significant figures.

Section A: multiple choice — 15 marks. Section B: structured — 17 marks.

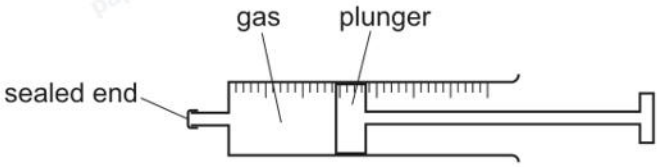
The number of marks is shown in brackets [ ] at the end of each question.

Section A · Multiple Choice (15 marks)

1. Which row describes the arrangement and the average separation of gas particles?

	arrangement of gas particles	average separation of gas particles
A	in fixed positions	less than in solids
B	in fixed positions	greater than in solids
C	in variable positions	less than in solids
D	in variable positions	greater than in solids

2. The diagram shows a sealed syringe containing a gas at a constant temperature.

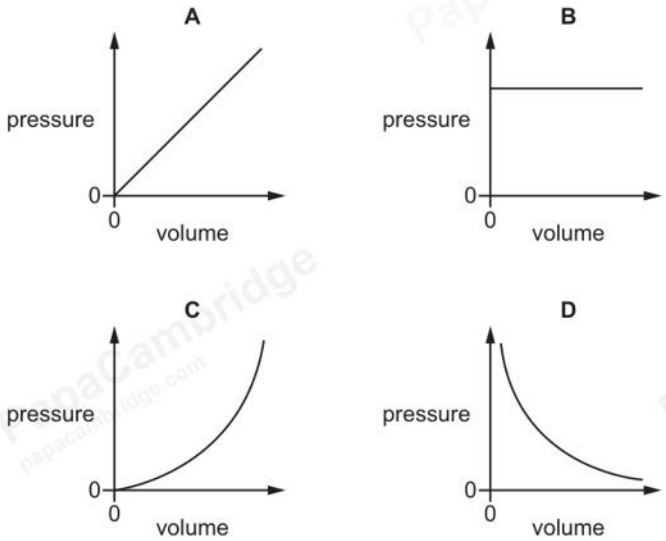


What happens to the gas particles and the pressure of the gas as the plunger is pushed further in?

	gas particles	pressure of the gas
A	hit the walls of the syringe less frequently	decreases
B	hit the walls of the syringe less frequently	increases
C	hit the walls of the syringe more frequently	decreases
D	hit the walls of the syringe more frequently	increases

3. The gas in a sealed container is compressed at constant temperature.

Which graph shows how the pressure of the gas changes with its volume?



4. Brownian motion is the random motion of particles.

In which states of matter is Brownian motion observed?

A gases, liquids and solids

B gases and liquids only

C gases and solids only

D liquids and solids only

5. 900 kJ of energy is transferred to a mass of water. The temperature of the water increases from 20 °C to 45 °C.

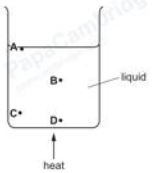
The specific heat capacity of water is 4200 J / (kg °C).

Which mass of water is heated?

A 0.0086 kg **B** 5.4 kg **C** 8.6 kg **D** 11 kg

6. A beaker of liquid is heated slowly so that the liquid evaporates.

During evaporation, from where do the more-energetic molecules leave the liquid?



7. What happens when the temperature of a liquid increases?

A The mass of the liquid increases, making the liquid less dense.

B The mass of the liquid increases, making the liquid more dense.

C The volume of the liquid increases, making the liquid less dense.

D The volume of the liquid increases, making the liquid more dense.

8. A warm black surface emits radiation. It is decided to increase the amount of radiation emitted.

Which changes increase the amount of radiation emitted?

1 making the surface hotter

2 changing the colour of the surface to white

3 decreasing the area of the surface

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 1 only

9. Which statement describes convection in a liquid?

A The particles expand and become less dense.

B The hot liquid becomes less dense and rises.

C The hot liquid becomes more dense and rises.

D The particles get further apart and fall.

10. Which thermal process can transfer energy through a vacuum?

A conduction

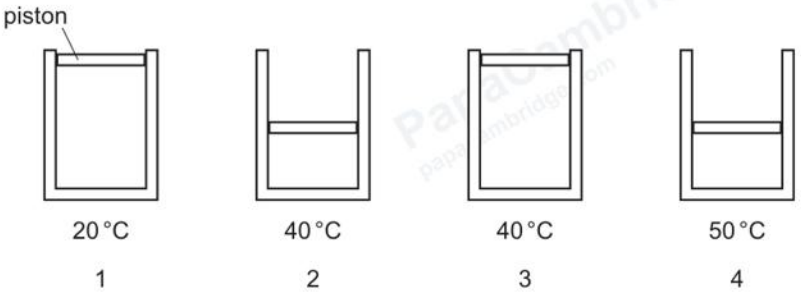
B convection

C evaporation

D radiation

11. The same mass of a gas is trapped in four identical cylinders by a piston that can move.

The diagrams show the samples of gas in different conditions of volume and temperature.



Which list gives the pressure of the gas in order from lowest to highest?

A $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$

B $1 \rightarrow 3 \rightarrow 2 \rightarrow 4$

C $4 \rightarrow 2 \rightarrow 3 \rightarrow 1$

D $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

12. A gas is trapped inside a cylinder which is sealed by a freely moving piston.

The piston is slowly pushed in so that the volume of the gas is reduced, but the temperature does **not** change.

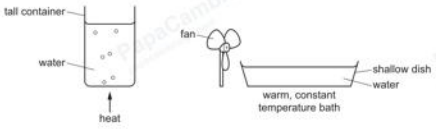
What happens to the kinetic energy of the gas particles and the rate at which they collide with the piston?

	kinetic energy of particles	**rate at which they collide with the piston**
A	no change	no change
B	no change	increases
C	increases	no change
D	increases	increases

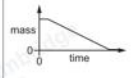
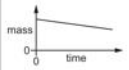
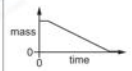
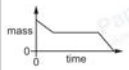
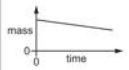
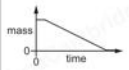
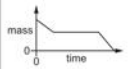
13. A student uses a tall container and a shallow dish for two experiments heating water.

The same mass of water at room temperature is used in each experiment.

The tall container is heated until the water boils and the shallow dish is kept warm at a constant temperature with a fan blowing across it.



Which row shows the correct graph for the mass of water remaining in each experiment?

	tall container	shallow dish
A		
B		
C		
D		

14. On a cold day, a shiny metal rod feels colder to the touch than a black plastic rod.

Which statement explains this observation?

A The metal rod is a better absorber of infrared radiation than the plastic rod.

B The metal rod is a better thermal conductor than the plastic rod.

C The metal rod is a worse absorber of infrared radiation than the plastic rod.

D The metal rod is a worse thermal conductor than the plastic rod.

15. A student suggests three processes by which thermal energy is transferred through space from the Sun to the Earth.

1 conduction

2 convection

3 radiation

Which processes are correct?

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 3 only

Question 1 · 8 marks

Air in a car tyre — gas pressure and particle structure

DO NOT WRITE IN THIS MARGIN

5 **(a)** Atmospheric pressure is 1.0×10^5 Pa. A volume of $1.02\,m^3$ of air at atmospheric pressure is pumped into a car tyre. The pressure of the air when it is inside the tyre is 2.6×10^5 Pa.

Calculate the volume of the air when it is in the car tyre.

volume = ... m^3 [3]

(b) **(i)** Describe the structure of a gas in terms of the separation, arrangement and motion of particles.

separation .....

.....

arrangement .....

.....

motion .....

.....

[3]

(ii) Explain, in terms of particles, why the internal energy of the gas decreases when the gas is at a lower temperature.

.....

..... [2]

[Total: 8]

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Question 2 · 9 marks

Heating water in a saucepan — specific heat capacity

4 Fig. 4.1 shows a stainless-steel saucepan being heated on an electric cooker. The saucepan contains water.

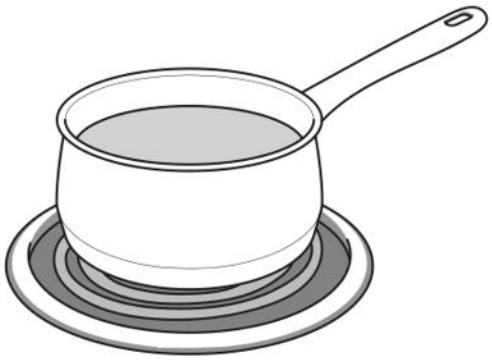


Fig. 4.1

(a) State what happens to the water particles as the water temperature increases.

..... [1]

(b) The saucepan contains 250 cm^3 of water. The specific heat capacity of water is 4200 J / (kg °C). The density of water is 1000 kg / m^3.

(i) Show that the mass of the water in the saucepan is 0.25 kg.

[2]

(ii) Calculate the energy required to increase the water temperature from 20 °C to 65 °C.

energy = ... [3]

(iii) The heater supplies enough power to heat the water in 39 s.
A student measures the time taken to heat the water as 115 s.

Suggest why the actual time taken to heat the water is longer. Assume that the student takes accurate measurements.

.....

..... [1]

(c) The stainless-steel saucepan is replaced with an aluminium saucepan of the same mass. It contains the same volume of water.

The specific heat capacity of stainless steel is 500 J / (kg °C).
The specific heat capacity of aluminium is 890 J / (kg °C).

Explain how using an aluminium saucepan will affect the time taken to heat the water.

.....

.....

..... [2]

[Total: 9]